

🏆 Smart India Hackathon 2025

Road Safety AI System

Real-time Hazard Detection + City-scale Digital Twin

🧠 YOLOv8 AI

🚌 100 Virtual Buses

🗺️ Delhi Routes

⚡ Live WebSocket

🚒 Emergency Pipeline

Domain

Smart Transport & Safety

Stack

Python · FastAPI · YOLOv8 · OpenCV

Scale

City → State → India-wide



Pothole Detected

conf 82% · BUS-023 · 2s ago



Accident Alert

🚒 CRITICAL · Emergency active



Traffic Light: RED

YOLOv8 · conf 92%

THE PROBLEM

India's Roads Are Dangerously Unmonitored

Manual inspection is slow, reactive, and geographically limited — costing lives and billions of rupees annually.

1.53L

ROAD DEATHS / YEAR

5 crore

POTHOLES ON INDIAN ROADS

₹1.5L Cr

ANNUAL ECONOMIC LOSS

70%

ACCIDENTS PREVENTABLE



Delayed Reporting

Potholes & road damage are reported days after appearing. Accidents rely on bystander calls.



No Real-time Visibility

Authorities have zero live awareness of road conditions across city networks.



Fragmented Response

Police, ambulance & road authority operate in silos — no unified command.

OUR SOLUTION

One Unified Platform for Smart Road Safety



AI Detection Engine

YOLOv8s + OpenCV CV pipeline detecting potholes, accidents, waterlogging, traffic lights in real-time.

5 hazard types

82–92% conf



City Digital Twin

100 virtual buses across 10 real Delhi routes, GPS movement, live Leaflet.js map with heatmap.

Real coordinates



Emergency Pipeline

Automatic escalation: Police → Ambulance → Road Authority → Resolved with live timeline.

Auto-dispatch



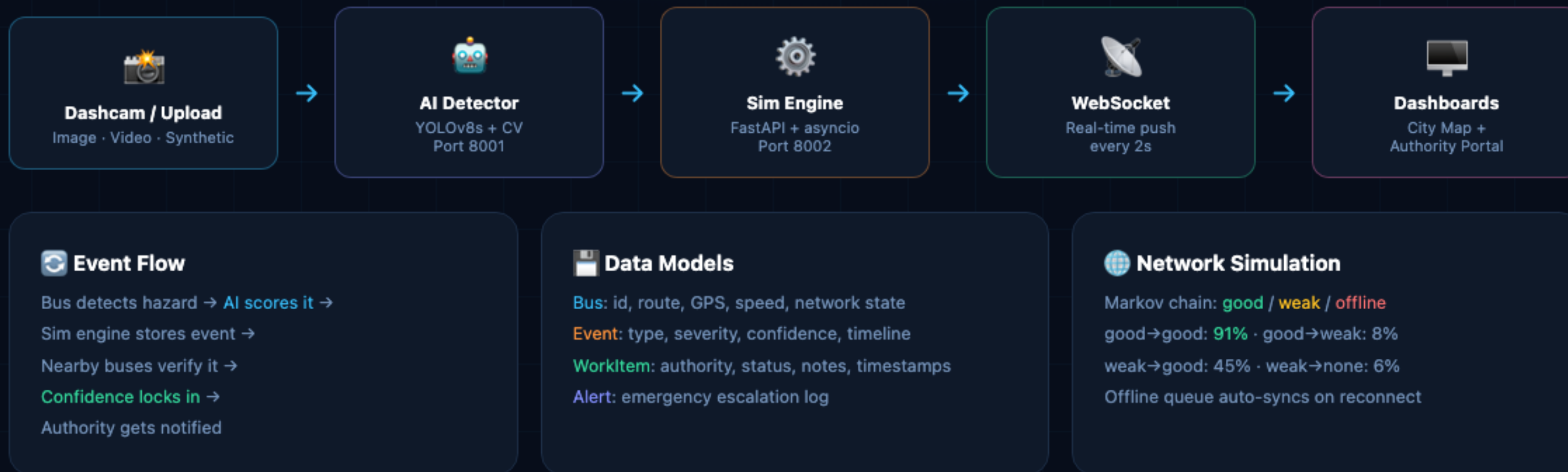
Authority Dashboard

Dedicated command center for each authority type — notifications, work tracking, completion stats.

4 authority types

"From pothole on road to authority response — fully automated, under 5 minutes."

How It All Connects



YOLOv8s + Computer Vision

Dual-layer Detection

 **Layer 1 — YOLOv8s (21.5 MB)**

 Vehicle / Accident Detection

COCO cls 2,3,5,7

 Traffic Light State (R/Y/G)

HSV analysis

 Pedestrian on Road

Lower 60% zone

 **Layer 2 — OpenCV CV Fallback**

 Pothole (adaptive threshold + dark pixel OR)

 Waterlogging (blue/brown HSV pool >15%)

 Multi-vehicle incident (overlapping colour blobs)

 **Accident Multi-signal Score**

Vehicle IoU overlap

x0.6

Person near vehicle

+0.35

High optical flow motion

+0.20

Dense vehicle cluster

+0.25

Threshold >0.45 → HIGH · >0.75 → CRITICAL

 **Detection Outputs**

✓ Annotated JPEG with bounding boxes

✓ Base64 encoded (JSON-safe)

✓ Per-detection: class, conf, bbox, severity

✓ Traffic density + advice

✓ Inject-to-map button → live simulation

100 Buses • 10 Routes Delhi in Real-time

10 Real Delhi Corridors

- | | |
|--|--|
|  North Ring Road |  East Delhi |
|  South Delhi |  Central Corridor |
|  West Delhi |  NH-48 |
|  Outer Ring North |  South Loop |
|  Airport Express |  Inner Ring |

Simulation Engine

-  Tick rate: 2s per step (adjustable speed)
-  GPS: Haversine + linear waypoint interpolation
-  Network: Markov-chain good/weak/offline
-  Offline queue: auto-syncs on reconnect
-  Events: 8 types, weighted probability
-  Verification: unverified → full (65m radius)

100

LIVE BUSES

8

EVENT TYPES

250

MAX EVENTS

8min

EVENT TTL

8 Road Event Types

- | | | |
|---|--|--|
|  Pothole 27% |  Road Crack 18% |  Waterlogging 12% |
|  Heavy Traffic 14% |  Accident 8% |  Roadblock 8% |
|  Fallen Tree 6% |  Animal 7% | |

Multi-Bus Verification

Crowd-sourced Confidence

1

First Bus Detects

Bus detects hazard, AI assigns confidence 60–98%. Event logged as **UNVERIFIED**.

2

Second Bus Passes

Any other bus within **65 metres** corroborates
→ status becomes **PARTIALLY VERIFIED**.

3

Third Bus Confirms

One more verification → **FULLY VERIFIED**.
Confidence now actionable for authorities.

4

Authority Notified

Verified events routed to the correct authority type. Work item created in their dashboard.

Why Multi-bus Verification?

- ✓ Eliminates single-sensor false positives
- ✓ Simulates crowdsourced ground truth
- ✓ Scales naturally — more buses = faster verification
- ✓ Works offline (syncs when network returns)

Verification Radius

65m

Haversine distance calculation between bus GPS and event coordinates. A passing bus within this radius automatically counts as a verification corroboration.

Automatic Escalation

Accident to Resolved — 6 min

	Critical Event Detected Bus AI flags: accident / fallen tree / roadblock with high/critical severity → Emergency protocol activated immediately.	T + 0s
	Police Unit Dispatched Delhi Police command center receives GPS-precise alert. Nearest unit auto-assigned. Status: police_notified .	T + 30s
	Ambulance En Route EMS dashboard shows incident card. Paramedics dispatched with precise coordinates. Status: ambulance_notified .	T + 90s
	Road Authority Alerted PWD / MCD gets work order for road clearance & repair. Status: authority_notified .	T + 150s
	Incident Resolved & Road Cleared Authority marks task complete on dashboard. Event archived. Heatmap updated. Full audit trail preserved.	T + 6min

 **Critical events: accident · fallen tree · roadblock**
Only high/critical severity events trigger emergency protocol — low severity goes to standard authority workflow.

Dedicated Portal for Every Authority Type



Delhi Police

Accidents, roadblocks, heavy traffic



EMS / Ambulance

Critical accident response only



PWD Road Authority

Potholes, cracks, waterlogging, trees



Municipal Corp

Waterlogging, animals, potholes



Live Stats Bar

Total assigned, New, In Progress, Completed with animated progress bar



Kanban Work Board

3-column board: New → In Progress → Completed with drag-to-update buttons



Live Toast Notifications

Pop-up alerts when new events are assigned, auto-refresh every 3s



Notes & Timeline

Officers add field notes, full event timeline shown per card



Filter & Search

Filter by Critical / High / Unverified / Emergency, search by event ID

5 Live Detection Scenes

Click → Annotated in seconds

SCENE 01



Pothole Road

Dark asphalt with 3 potholes + road cracks. CV adaptive threshold + dark-pixel OR.

4 potholes

conf 74–82%

SCENE 02



Traffic Light (RED)

Two traffic lights on poles. YOLOv8 detects both, HSV confirms RED state.

2 traffic lights

conf 88–92%

SCENE 03



Traffic Accident

3 overlapping vehicles (different colours). CV detects cross-band bbox overlap → collision.

1 accident

conf 71% HIGH

SCENE 04



Waterlogging

Large blue/muddy pool covering >40% of road. HSV pool detector + stranded car.

waterlogging

conf 91% HIGH

SCENE 05



Heavy Traffic

15 cars in 3 lanes + red traffic light. YOLO detects light, CV detects road texture anomalies.

1 traffic light

4 potholes



Upload Your Own

Drag & drop any road photo → instant AI annotation + inject to live map

Any image format

SCALABILITY

City → State → India-wide

SCALE MODE	BUSES	EVENTS/MIN	API RPS	ALERT LAG
City (100)	100	~10	5	800ms
State (1K)	1,000	~100	50	1.5s
India (10K)	10,000	~1,000	500	4.2s

India-wide Extrapolation

-  BEST Mumbai alone: 3,400 buses
-  DTC Delhi: 7,000+ buses
-  All India state RTCs: ~1.5 lakh buses
-  Cities covered: 500+ with GPS fleet

Horizontal Scaling

-  Asyncio event loop — non-blocking
-  WebSocket fan-out to N clients
-  Scale parameter multiplies all metrics
-  Deploy on cloud — Redis + Kafka ready

Future: Real IoT Integration

- Replace simulated buses with real GPS trackers
- Plug in dashcam feeds from bus CCTV
- OpenStreetMap + GTFS route data
- SMS/push alerts to civic workers

TECHNOLOGY STACK

Built With Production-grade Tools



YOLOv8s

Object detection



OpenCV

CV / image processing



FastAPI

REST + WebSocket API



asyncio

Non-blocking sim loop



Leaflet.js

Interactive map



leaflet.heat

Incident heatmap



Python 3.11

Core runtime



NumPy

Array / matrix ops



Ultralytics

YOLO model wrapper



WebSockets

Real-time push



CartoDB Tiles

Dark map tiles



Haversine

GPS distance calc

~1,800

LINES OF PYTHON

~2,600

LINES OF HTML/JS

3

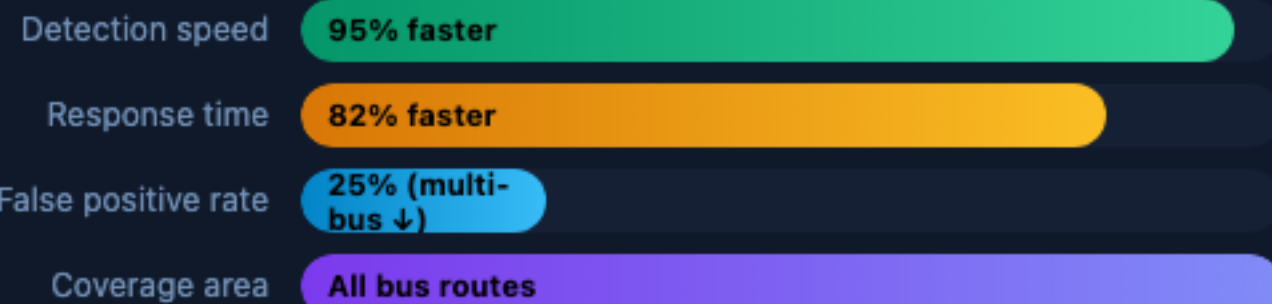
HTML DASHBOARDS

15+

API ENDPOINTS

Saving Lives Through Data-Driven Action

Demonstrated Impact



Cost Advantage

Uses **existing bus GPS + dashcams** — zero new hardware

Model size: **21.5 MB** — runs on Raspberry Pi

Open source stack — **₹0 licensing**

1 server handles 100+ buses real-time

Phase 2 Roadmap



Real IoT Integration

Live GPS + dashcam from actual BEST / DTC buses



Fine-tuned Pothole Model

Custom YOLOv8 trained on IDD / BDD100k pothole dataset



Mobile App for Officers

Field worker app with offline detection capability



Cloud Deploy (Redis + Kafka)

Production-grade message queue for 10K+ buses



Thank You

Road Safety AI System — Smart India Hackathon 2025

 YOLOv8s Detection

 100-Bus Simulation

 Emergency Pipeline

 Authority Dashboard

 Launch City Simulation

 Authority Dashboard

5

HAZARD TYPES

100

VIRTUAL BUSES

4

AUTHORITIES

6 min

FULL RESPONSE

10K

BUS CAPACITY